

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Find the particular solution of the initial value problem:

$$x \frac{dy}{dx} = 2y + x^3 \sin x \cos x, \quad y(\pi) = 2.$$

(First transform the differential equation into the standard form of linear first-order differential equation if it is not.)

Solution: standard form:

$$\frac{dy}{dx} - \frac{2}{x}y = x^2 \sin x \cos x$$

$$P(x) = -\frac{2}{x}, \quad Q(x) = x^2 \sin x \cos x$$

Integrating factor

$$f(x) = e^{\int P(x) dx} = e^{\int -\frac{2}{x} dx} = e^{-2 \ln |x|} = |x|^{-2} = \frac{1}{x^2}$$

We get

$$\frac{d}{dx} \left[\frac{1}{x^2} y \right] = \frac{1}{x^2} \cdot x^2 \sin x \cos x$$

$$\frac{y}{x^2} = \int \sin x \cos x dx$$

$$\frac{y}{x^2} = \frac{1}{2} \sin^2 x + C$$

$$\Rightarrow y = x^2 \left(\frac{1}{2} \sin^2 x + C \right) = 2$$

$$y(\pi) = 2 \Rightarrow \pi^2 \left(\frac{1}{2} \sin^2 \pi + C \right) \Rightarrow \pi^2 C = 2 \Rightarrow C = \frac{2}{\pi^2}$$

$$\text{Thus } y(x) = x^2 \left(\frac{1}{2} \sin^2 x + \frac{2}{\pi^2} \right)$$